Virtual Reality to Improve Autism Social Skills with a "Gotong Royong" Context

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Abstract

This research explores the use of Virtual Reality (VR) technology, specifically the aframe.io framework, to improve social skills in children with autism within the Indonesian cultural context of "Gotong Royong." Addressing challenges associated with Autism Spectrum Disorder (ASD), the study focuses on sustainable social development goals. Eight participants, aged 10 to 18, with mild autism engaged in VR-based activities like virtual communal clean-ups. Utilizing the Technology Acceptance Model (TAM), the study evaluates factors such as Perceived Ease of Use (PEOU) and Perceived Usefulness (PU) in the context of social skill enhancement. Results indicate positive perceptions, with average scores of 3.5 for PEOU and 3.8 for PU, suggesting moderate ease of use and significant usefulness. The Attitude toward Using (ATU) average score of 3.5 reflects a positive mindset, while the Behavioral Intention to Use (BITU) average score of 3.8 suggests a strong motivation to integrate the technology into daily routines. However, the Actual System Use (ASU) score of 2.8 highlights the need for further evaluation and improvement. In conclusion, this research demonstrates the potential of VR technology within a "Gotong Royong" context to enhance social skills in children with autism. The findings contribute to refining VR interventions, addressing social challenges in individuals with autism, and advancing toward sustainable social development goals.

Keywords: *Virtual Reality, Autism, Gotong Royong, Technology Acceptance Model, Social Skills*

A. Introduction

Children are the next generation and the hope of the family and the country. Children have special rights [1]. But this can be interfered by one factor, namely autism.

Autism Spectrum Disorder (ASD) is defined by social and communication problems, along with repetitive behavioral patterns and restrictions that can vary in severity between individuals [2]. There are three severity level, namely level 1, which is low; level

2, which is moderate; and level 3, which is high [3]. Symptoms of autism are characterized by minimal social communication and social interaction in various contexts, along with restricted and repetitive patterns of behavior and interests or activities [4]. ASD is by heterogeneous neurodevelopmental disorders caused by genetic and environmental factors [5]. Based on average figures and prevalence, people with the above symptoms are estimated to be 1 out of 100 children in the world {1}.

Based on the explanation above, autism can cause socialization deficits in humans {2}. Meanwhile, according to Aristotle, humans are social creatures (*Zoon Politicon*), so social skills are fundamental for every human being, including autistic people. If they don't develop social skills until adulthood, autistic people have no hope for the future [6].

That is one of the obstacles to the realization of the SDGs. There are several SDG goals that are relevant to this issue. In SDG 3, the disadvantage of autistic people is caused by disturbed mental health. In SDG 4, individuals with intellectual disabilities have higher rates of dropping out of school, lack of educational degrees, and school refusal due to autistic social interaction problems [7], [8]. Because of this, in SDG 10, autistic people do not receive equality [9]. That is why a solution is needed that can be one way to solve problems related to {3}{4}.

Technological advances have the potential to be a solution to overcome these problems. Virtual Reality (VR) is one of them. VR is used in various fields [10]–[13]; [14]–[16]; [17]; one of them is disability {5}. VR is a computer-generated representation of an environment that simulates physical presence in the real world or in the virtual world [18]. In this research, the type of VR used is immersive VR because the resulting representation feels more realistic, so the product users feel like they have truly entered the VR world {6}.

VR can be a medium for developing autistic social skills in cyberspace. That is because VR has the potential to control and manipulate different features of social situations, helping students with autism better adapt to and solve them [19]. Apart from that, according to the results of interviews conducted with autism teachers at SLB N 1 Bantul, children with autism are interested in visual forms such as colorful pictures and videos. The above proves that VR is possible for use by autistic people.

There are five studies [20]–[24] that are relevant to this research, with the research theme of developing VR simulation products as a medium for developing social skills in autism.

These five studies show that VR is interesting and can significantly develop social skills. However, VR needs to be developed so that more and more communities are aware of it, such as through scenario development, the development of a set of goals for practice and pedagogy regimens equivalent to standard training, and dynamic development in aligning the type, sequence and social stimulation of training activities oriented towards bond design based on the involvement and performance of each learner.

This research complements previous research related to VR to {7} improve the social skills of people with autism. Until now, no research related to this has raised the theme of local wisdom related to social skills. In this research, VR was used to improve the social skills of autistic children, highlighting the context of local Indonesian wisdom, namely mutual cooperation as a form of connection with social skills. Mutual cooperation is a form of social skill because in mutual cooperation, togetherness must based on social action and social solidarity [25]. *Gotong royong* is a noble Indonesian culture that is rooted in the values of *Pancasila*, making it the basis of social solidarity among Indonesian people [26]. The value of mutual cooperation needs to be preserved through VR, which simulates this.

Research Hypothesis: The use of Virtual Reality (VR) technology in the context of '*Gotong Royong*' will result in a significant increase in the social skills abilities of individuals with autism, as measured through various parameters such as social interaction, communication skills, and understanding of the concept of '*Gotong Royong*'.

B. Method and Experimental Details

1. Participants and Setting

Table 1. List of autism participants

No	Participants	Age	Gender	Level
1	Satriadi Cahyo P	10	Male	Mild
2	Jenicka Melody K	12	Female	Mild
3	Rizky Ramadhani	12	Male	Mild
4	Adinda Mei Risti	13	Female	Mild
5	Saka Danadyaksa S	16	Male	Mild
6	Rizka Putri L	17	Female	Mild
7	Affandi Akbar	18	Male	Mild
8	Rizal Julianto	18	Male	

The participants are from Special School (SLB) N 1 Bantul, encompassing students with educational backgrounds ranging from elementary school to high school. The respondents consist of 5 male and 3 female children, with ages ranging from 10 to 18 years. All participants have a mild level of autism severity. The selection process was based on their ability to follow instructions, ensuring that participants have the capacity to engage with the instructional aspects of the study.

2. Intervention: VR-Based Social Skills (gotong royong) Training

Figure 1. VR view of area playground V-TORO for autism user

This research employs the aframe.io web framework as a platform to create visual experiences in HTML-based virtual reality (VR). The focus lies on tasks involving social interaction integrated into everyday scenes within a simulated VR environment, such as collaboratively cleaning up trash. Social interaction tasks engage two users or operate in multiplayer mode, implemented based on the concept of cooperation in a specific task, namely, disposing of bottles together. The facilitator, in this case, the researcher, guides these tasks using the V-TORO concept. Users view an initial screen displaying a field with various elements, including a house, tractor, straw, plants, and key objects such as trash bottles and bins. Users navigate and control their perspective using VR controllers. The V-TORO usage scenario includes adaptation to prevent motion sickness, environmental identification through facilitator questions, and the execution of a collaborative scenario involving disposing of trash together by two users side by side.

Figure 2. Prespective of View of V-TORO from user when playing together with other player in multiplayer mode.

Figure 3. Autism Children when Playing V-TORO

3. Development Model

The ADDIE (Analysis, Design, Development, Implementation, and Evaluation) instructional design framework is a systematic approach to creating effective learning experiences. In the analysis phase, the designer identifies learning needs and goals, followed by the design phase where instructional strategies and materials are outlined. The development phase involves creating the actual instructional content, while the implementation phase focuses on delivering the instruction to the target audience. Finally, the evaluation phase assesses the effectiveness of the instruction, providing insights for improvement. This iterative process allows for continuous refinement and enhancement of the learning experience.

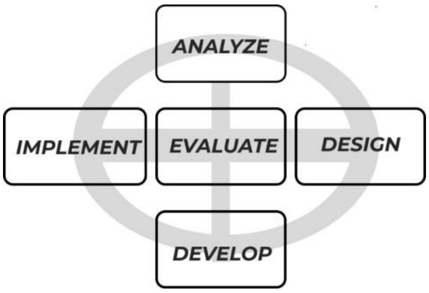

Figure 4. ADDIE model diagram

4. Instrument

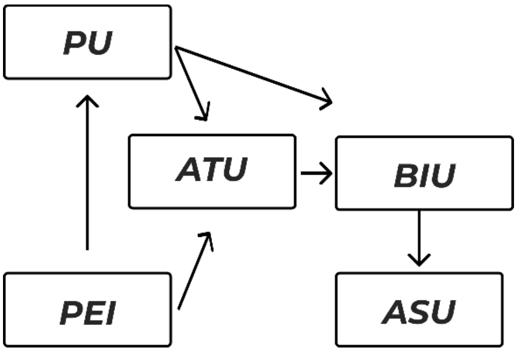

Figure 5. TAM model

The utilization of the Technology Acceptance Model (TAM) as the research instrument for this study is grounded in the need to comprehensively understand and evaluate the acceptance and adoption of virtual reality (VR) technology within the specific context of enhancing social skills in individuals with autism.

TAM, a well-established theoretical framework, offers a systematic approach to examining the factors influencing users' acceptance of technology. In the context of autism, where social skills development is a critical aspect, TAM provides a structured lens to assess key elements such as Perceived Ease of Use (PEOU) and Perceived Usefulness (PU), which are vital in determining the feasibility and effectiveness of integrating VR technology for social skill improvement.

The choice of TAM as the instrument allows researchers to delve into the attitudes and intentions of individuals with autism towards using VR technology, exploring their perceptions of ease of use and the perceived utility of the technology in the specific context of social skill enhancement. By assessing attitudes and intentions, TAM facilitates a deeper understanding of the potential success and user acceptance of the proposed VR intervention.

Moreover, TAM's inclusion aligns with the research focus on a "Gotong Royong" context, emphasizing collaboration and communal engagement. The model aids in examining not only the individual user's perspective but also the collective social dynamics that may influence the adoption of VR technology within a communal setting, shedding light on how the technology may be embraced and integrated into shared social activities.

In summary, the incorporation of TAM as the research instrument is justified by its capacity to systematically evaluate the crucial factors influencing the acceptance of VR technology among individuals with autism in a "Gotong Royong" context. It provides a structured framework for assessing the perceived ease of use, usefulness, attitudes, and behavioral intentions, contributing valuable insights to inform the development and implementation of effective VR interventions for enhancing social skills in individuals with autism.

Table 2. Variable and TAM indicator

Variabel	Indicator	Symbol
	Easy to use	PEOU1

Perceived Ease of Use	Easy to understand	PEOU2
	Ease of achieving goals	PEOU3
Perceived Usefulness	Makes work easier	PU1
	Improve the performance	PU2
	Helpful	PU3
Attitude Towards Using	Acceptance	ATU1
	Rejection	ATU2
	Enjoying the Usage	ATU3
Behavioral Intention to Use	Plan to keep using in the future	BITU1
	Motivation to keep using	BITU2
	Using in any condition	BITU3
Actual System Use	Match with procedures	ASU1
	Honesty in use	ASU2
	Duration of usage	ASU3

C. Result and Discussion

Table 3. Respondents Data from TAM Questionare.

Res	**PEOU**	**PU**	**ATU**	**BITU**	**ASU**
1	4	5	4	5	3
2	3	4	3	4	2
3	5	3	5	3	4
4	4	5	4	5	3
5	2	4	2	4	1
6	3	2	3	2	3
7	5	4	5	4	5
8	4	3	4	3	4
9	3	5	3	5	2
10	2	3	2	3	1
Total	35	38	30	38	2,8
AVG	3,5	3,8	3,0	3,8	2,8

1) Perceived Ease of Use (PEOU):
The Perceived Ease of Use (PEOU) average score of 3.5 suggests that respondents generally find the technology fairly easy to use. This indicates a moderate level of perceived ease in interacting with the system. While not extremely high, the score implies that the users don't encounter significant challenges or complexities when using the technology. It may be considered a positive indicator, but further exploration into specific aspects of user interface and usability could provide insights into potential improvements.

2) Perceived Usefulness (PU):
With an average score of 3.8 for Perceived Usefulness (PU), respondents tend to view the technology as quite useful. This suggests that users see value and benefits in utilizing the technology. The higher score indicates a positive perception of the technology's utility, implying that it aligns well with users' needs and expectations. The results reflect a favorable attitude toward the practical benefits offered by the technology, which is crucial for user acceptance.

3) Attitude toward Using (ATU):
The average Attitude toward Using (ATU) score of 3.5 indicates that respondents generally hold a positive attitude toward using the technology. This positive inclination suggests that users are likely to approach the technology with optimism and a favorable mindset. A balance between perceived ease of use and perceived usefulness contributes to this positive attitude, indicating that users are receptive to

engaging with the technology in a constructive manner.

4) Behavioral Intention to Use (BITU):
The average Behavioral Intention to Use (BITU) score of 3.8 implies that respondents tend to have a positive intention to use the technology. This suggests a strong likelihood that users are motivated to incorporate the technology into their regular routines or activities. A higher intention to use reflects a positive predisposition toward adopting the technology in practical scenarios, reinforcing the idea that users perceive meaningful benefits and value in its application.

5) Actual System Use (ASU):
The average Actual System Use (ASU) score of 2.8 indicates the observed level of technology use. This score suggests that, on average, users are engaging with the technology at a moderate level. It's essential to evaluate whether this aligns with initial expectations or if there are areas that require improvement. A score below expectations may prompt further investigation into factors influencing actual usage, such as usability issues, training needs, or other user-related considerations.

D. Conclusion

The utilization of virtual reality (VR) technology to enhance social skills in children with autism within the context of "Gotong Royong" has been evaluated through key parameters. The perceived ease of use (PEOU) indicates that respondents generally find the technology fairly easy to use, suggesting a user-friendly interface. Perceived usefulness (PU) results show that respondents perceive the VR technology as quite useful, emphasizing its potential benefits. Attitude toward using (ATU) reflects a positive disposition among respondents regarding the technology's usage. Additionally, behavioral intention to use (BITU) demonstrates a positive inclination among respondents to incorporate the technology into their routines. However, the average level of actual system use (ASU) is measured at 2.8, suggesting a need for evaluation to align expectations and identify potential areas for improvement. This research underscores the promising role of VR technology within a 'Gotong Royong' framework for enhancing social skills in children with autism, with room for further refinement in its implementation.

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material

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